

Laboratory Theory Examination

MSc Physics Sem 4 2026

Time Allowed: 3 Hours **Maximum Marks:** 30

Instructions:

- Attempt **all** questions. All questions carry equal marks.
 - Use of non-programmable scientific calculator is permitted.
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- Q1.** Which laser is used in the speed of light experiment in the lab? Describe the path of the light ray in the experiment and derive the formula to calculate the speed of light.
- Q2.** What is the origin of Stokes and anti-Stokes lines in Raman spectra? Which one would have more intensity and why?
- Q3.** Explain the significance of Stefan's law. What do you mean by a thermocouple? Which thermocouple is used in the lab during the experiment?
- Q4.** Which substance is used in the ESR experiment and why? Explain the working and theory behind the experiment.
- Q5.** What are the different motions possible in a gyroscope? How are they interrelated with each other?
- Q6.** State Bragg's law and explain how it is conceived in the experiment.
- Q7.** Explain the construction and working of a GM counter. What is dead time in a GM counter?
- Q8.** Can we measure gamma rays using a GM counter? Explain why. What are the different peaks observed in the gamma-ray spectrum of a nucleus? Arrange them in order of their energies.
- Q9.** Derive the formula for the observed frequency when both the source and the observer are moving relative to each other.

- Q10.** Why is a transverse voltage observed in some metals in the absence of an applied external magnetic field? Why do some metals have a positive Hall coefficient? Give an example.
- Q11.** Which gas is used in PQMS and why? Explain the significance of the χ vs. L graph. What are the different geometries in the four-probe method? Explain which one is preferred over the other.
- Q12.** What is the working principle of STM? What are the advantages and disadvantages of different imaging methods in STM?

— *End of Question Paper* —

Laboratory Theory Examination

MSc Physics Sem 2 2026

- Q1.** How to identify capacitor, inductor and terminals of BJT using Multimeter
- Q2.** Derive the ripple factor for half wave and full wave rectifiers
- Q3.** Frequency response of an integrator at low and high frequency output, and its Bode plot
- Q4.** Frequency response and Bode plot of a CE (Common Emitter) configuration amplifier
- Q5.** Limitations of a half subtractor and its diagram using XOR and AND gates
- Q6.** Op-amp IC diagram and its explanation
- Q7.** JFET pinch-off voltage and how to determine it experimentally
- Q8.** Limitations of an RS flip-flop and how JK improves it; limitations of a JK flip-flop
- Q9.** When does a Common Collector configuration become an Emitter Follower
- Q10.** What is resonance, and how do independent variations in resistance and capacitance affect an LCR circuit
- Q11.** Minimum number of phase-shifting RC stages required, and does a balanced Wien bridge give sustained oscillations
- Q12.** Common Emitter and Common Base output characteristics
- Q13.** I-V characteristics of an ideal PN junction, and the significance of knee voltage in a practical PN junction